SimCSE Reproduction — Final Scientific Synthesis

Scope: validated implementation, reduced-data scaling, controlled ablations, alignment/uniformity analysis, and discrepancy analysis.

1. Executive conclusion

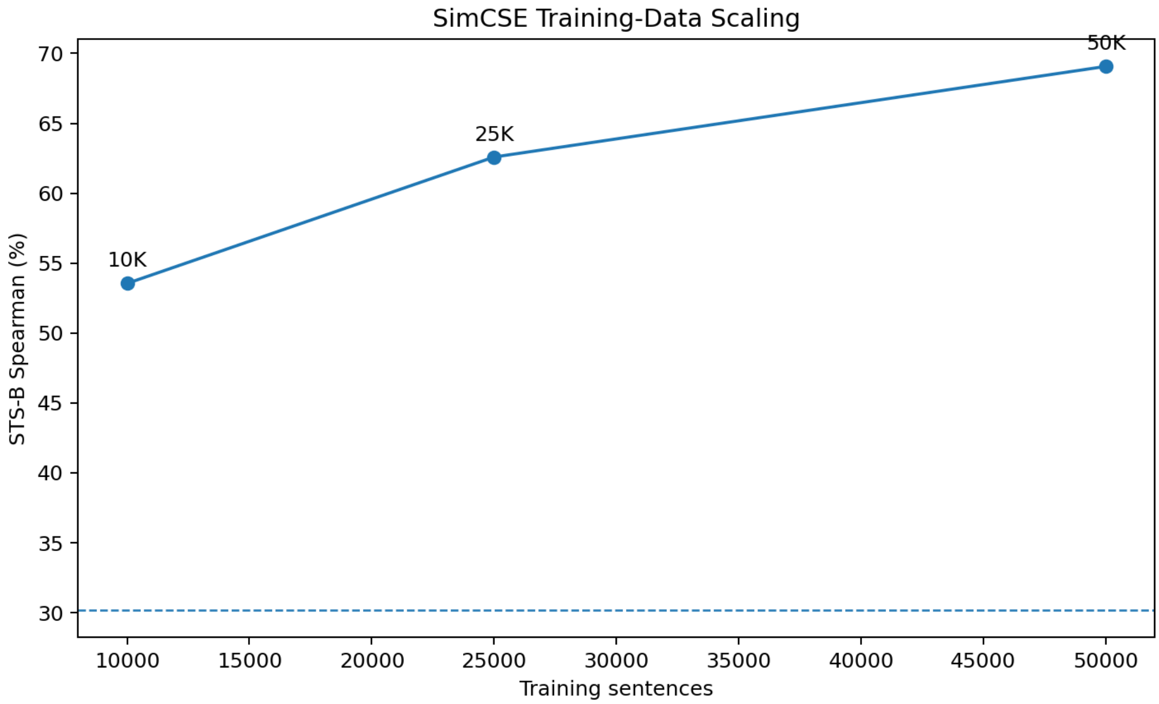
The project validates the core implementation of unsupervised SimCSE and completes the planned reduced-data scaling study, controlled dropout/temperature ablations, and representation-geometry analysis. The evidence supports a mechanism-level interpretation in which independent dropout creates useful positive-view variation and strong downstream performance depends on a useful alignment–uniformity balance.

2. Main results

Experiment	Size	STS-B
Raw BERT CLS	0	30.18%
SimCSE	10K	53.55%
SimCSE	25K	62.59%
SimCSE	50K	69.07%
No dropout	50K	60.21%
Fixed mask 0.10	50K	41.07%
Dropout 0.20	50K	76.38%
Temperature 0.01	50K	68.88%
Temperature 1.00	50K	58.33%
Official pretrained, local STS-B	—	82.12%

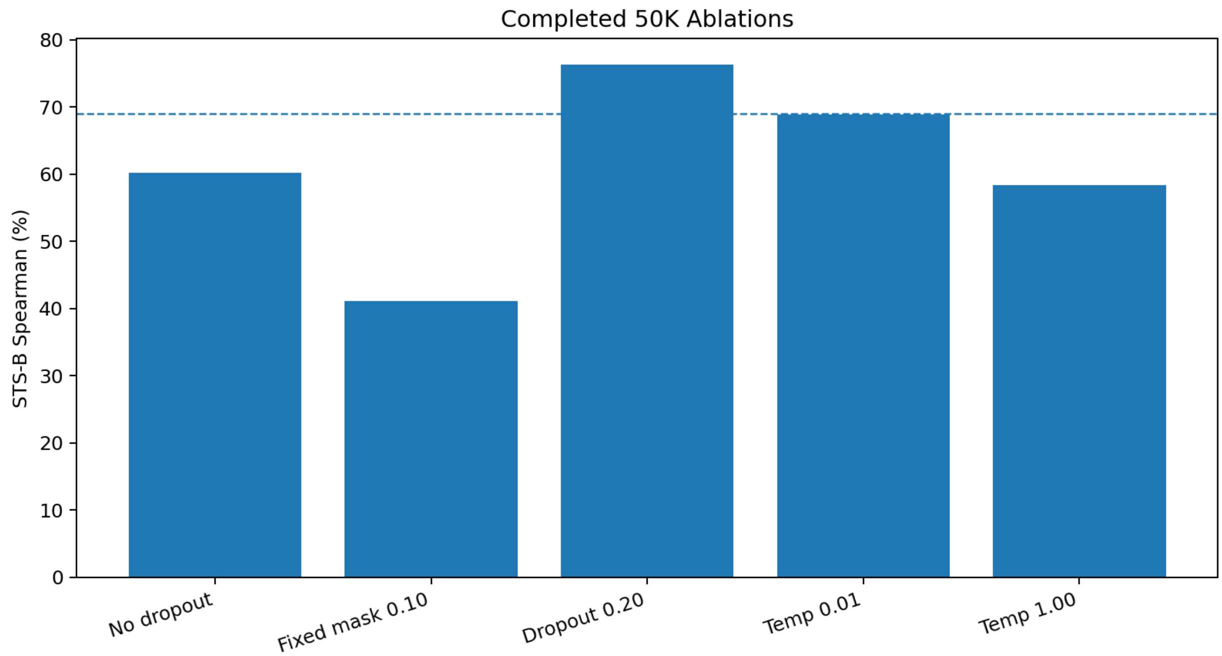
3. Training-data scaling

The reduced-data curve is monotonic: 53.55% at 10K, 62.59% at 25K, and 69.07% at 50K. The marginal gains fall from +9.04 percentage points to +6.48 percentage points. This is an observed diminishing-return pattern over the tested range, not a universal scaling law.


SimCSE Training-Data Scaling
STS-B Spearman (%)
Training sentences
10K
25K
50K
30
35
40
45
50
55
60
65
70
10000
15000
20000
25000
30000
35000
40000
45000
50000

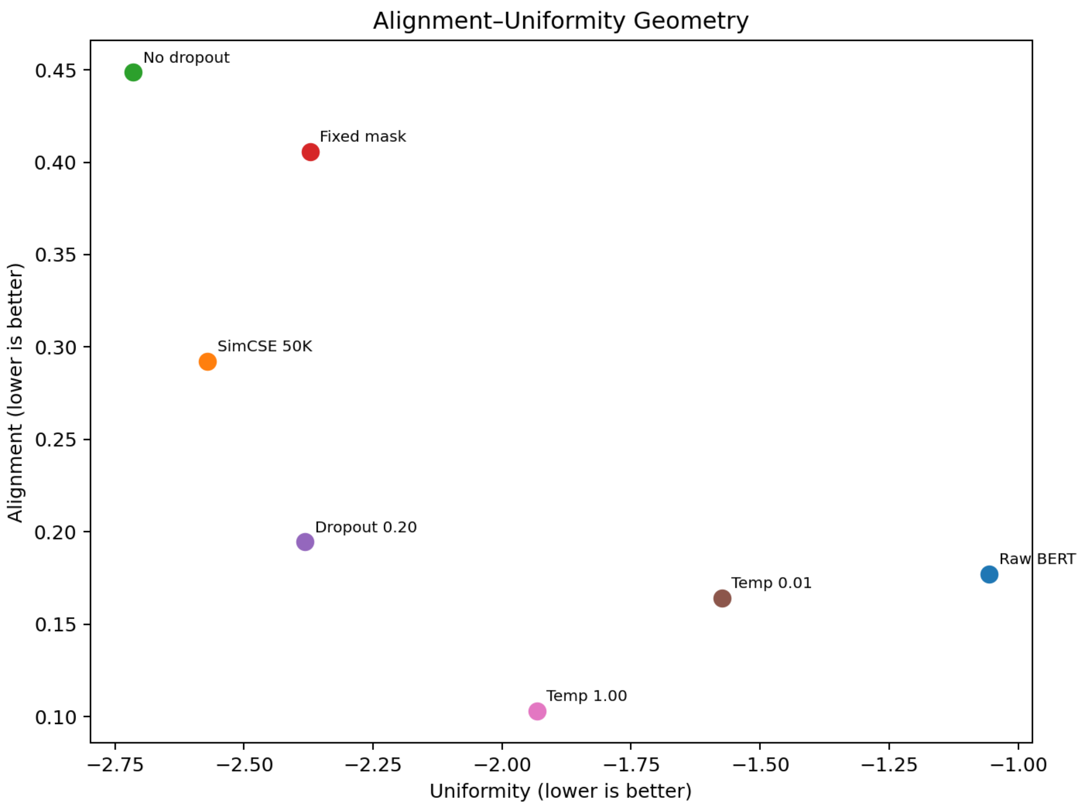
4. Controlled ablations

Independent stochastic dropout is the strongest mechanism-level finding. Removing dropout reduces STS-B to 60.21%; sharing the dropout mask reduces it further to 41.07%; standard dropout at 0.20 reaches 76.38%. Temperature also matters substantially: 0.01 gives 68.88% and 1.00 gives 58.33%.



5. Representation geometry

Experiment	STS-B	Alignment	Uniformity
raw_BERT	30.18%	0.1771	-1.0560
SimCSE_50K	69.07%	0.2924	-2.5705
no_dropout	60.21%	0.4486	-2.7144
fixed_mask_0.10	41.07%	0.4057	-2.3718
dropout_0.20	76.38%	0.1946	-2.3814
temperature_0.01	68.88%	0.1642	-1.5738
temperature_1.00	58.33%	0.1030	-1.9329



The geometry results show why neither metric should be optimized in isolation. No-dropout achieves very strong uniformity (-2.7144) but only 60.21% STS-B. Temperature 1.00 achieves the strongest alignment (0.1030) but only 58.33% STS-B. The strongest local STS-B result, dropout 0.20, combines good alignment (0.1946) with substantially improved uniformity (-2.3814). With only seven geometry conditions, the observed correlations are descriptive rather than inferential.

6. Reference and discrepancy interpretation

The published 76.25% SimCSE BERT-base result is the average across seven STS tasks, not an STS-B-only score. The paper reports STS-B separately at 76.85%. Therefore the local 76.38% STS-B result must not be described as beating the paper's 76.25% figure. The official pretrained checkpoint's locally measured 82.12% is an STS-B validation reference and is likewise not numerically equivalent to the seven-task average.

The official repository also evaluates during training and saves the best checkpoint, whereas the local reduced-data experiments evaluate the resulting epoch checkpoint. The local study additionally uses 10K/25K/50K Wikipedia subsets because full 1M-sentence CPU training was computationally impractical. These are protocol/resource differences, not evidence of a broken core implementation.

7. Final scientific claim

We reproduced and validated the core unsupervised SimCSE training implementation and characterized its learning behavior, mechanism-level ablations, and representation geometry under a resource-constrained reduced-data training regime.

8. Final limitations

• The main local training study is reduced-data because of CPU constraints.

• The 1M-sentence run was launched but not completed.

• The local training protocol does not reproduce every aspect of official best-checkpoint selection.

• Completed reduced-data models were locally evaluated on STS-B rather than the paper's seven-task average.

• Geometry conclusions are descriptive because only seven completed geometry conditions are available.